# Excel Data Analytics with Power BI

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**Project:** Company Auditing

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# Part A: Power BI Data Model

The Power BI data model integrates transaction and reference tables to support audit analysis across the revenue, purchasing, inventory and general ledger cycles. Relationships allow transactions to be examined using relevant customer, supplier, employee, product and account information.

In the revenue cycle, SalesTable is linked to CustomersTable, BranchesTable and ProductsTable through their respective identification fields. These many-to-one relationships enable sales analysis by customer, branch and product. SalesTable and CreditNotesTable are also connected to InvoiceLookup through invoice identifiers. This arrangement supports matching credit notes to original invoices, identifying multiple credits and investigating potential revenue cut-off issues.

For purchasing, PaymentsTable is connected to SuppliersTable through Supplier_ID. This enables analysis of repeated supplier invoice payments and payments to suppliers currently recorded as unapproved. Payment processing and approval can also be examined using relationships between PaymentsTable and EmployeesTable. The processor relationship is active, while the separate approver relationship is inactive to avoid competing filter paths.

For inventory, ProductsTable is linked to InventoryTable through Product_ID. This supports calculations of inventory value at recorded cost and identification of damaged and obsolete stock.

In the general ledger cycle, GLTable is related to TrialBalanceTable through Account_Code and to AccessTable through User_ID. These connections support account-level analysis and investigation of journal processing controls. The relationship between GLTable and BranchesTable is inactive so branch-specific general ledger analysis requires appropriate relationship handling.

A significant data-quality issue was identified in SalesTable: five invoice IDs (INV14991–INV14995) appear twice with conflicting customer or invoice-date information. Consequently, Invoice_ID is not a unique key in the sales data. A separate InvoiceLookup table was used to support relationships with credit notes. However, this does not resolve the underlying ambiguity, so invoice-level matching and credit percentages require additional verification.

Most many-to-one relationships use single-direction filtering from reference tables to transaction tables. The one-to-one relationships between EmployeesTable and AccessTable, and between ProductsTable and InventoryTable, use bidirectional filtering to support analysis across their matched records. Inactive relationships, including the general ledger branch relationship, help avoid ambiguous filter paths.

Overall, the model supports consistent calculations and cross-table audit testing. However, analytical exceptions are indicators of risk rather than proof of error or fraud. Supporting documents and authorisation records must be examined before drawing final audit conclusions.




*Figure 1: Power BI Data Model and Table Relationships*

# Part B: Audit Risk Identification

## B.1: Summary of Identified Audit Risks

| Risk | Audit cycle | Assertion(s) affected | Evidence from Power BI |
|---|---|---|---|
| Multiple credit notes against original invoices | Revenue/AR | Accuracy, valuation | Five invoices received two credit notes each, totalling $28,489.64 (42.02% of their combined original sales). Three invoices had credit ratios above 50%. |
| Post-year-end credit notes relating to FY2026 sales | Revenue/AR | Cut-off, accuracy | July 2026 credit notes totalling $187.46K relate to FY2026 sales invoices. |
| Potential duplicate supplier payment | Purchasing/AP | Occurrence, accuracy | Supplier S0012, invoice SUP5555, has two payments of $32,000 (PAY01995 and PAY01996), dated two days apart. |
| Payments to unapproved suppliers | Purchasing/AP | Occurrence, accuracy | Eighteen payments totalling $800,791.56 were made to suppliers S0233 and S0241, both currently marked unapproved. |
| Damaged inventory potentially requiring a write-down | Inventory | Valuation | Damaged inventory has a recorded cost of $2M, compared with total inventory of $57.4M. |

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| Obsolete inventory potentially requiring a write-down | Inventory | Valuation | Inventory flagged obsolete has a recorded cost of $227.46K. |
| Journal entries posted and approved by the same user | General Ledger/Journals | Occurrence, accuracy, classification | 39 journal entries have matching User_ID and Approver_ID values. |
| Users with both payment processing and payment approval access | Control Environment (User Access) | Occurrence, accuracy | Five users (U117, U051, U025, U018 and U013) have both Process_Payment = Yes and Approve_Payment = Yes. |
| Users with supplier creation and payment access | Control Environment (User Access) | Occurrence, accuracy | Two users (U117 and U018) have Create_Supplier = Yes and also have payment processing and approval access. |
| Post-termination system access | Control Environment (User Access) | Occurrence, accuracy | Two users (U117 and U118) show last-login activity after termination and are flagged 'After Termination'. |
| Reused sales invoice IDs with conflicting details | Revenue/AR | Accuracy, occurrence, cut-off | INV14991 – INV14995 each appear twice. The records contain different customers and/or invoice dates, indicating potential invoice-ID integrity problems. |

*Table 1: Summary of Identified Audit Risks Across Five Audit Cycles*

## B.2: False-Positive Investigation

| Initially suspicious item | Investigation evidence | Conclusion |
|---|---|---|
| Repeated supplier invoice number SUP58932 | PAY01239 relates to supplier S0088, purchase order PO3767, and $33,906.75. PAY00002 relates to supplier S0206, purchase order PO5754, and $27,691.79. The payments also have different bank accounts and dates. | False positive in an invoice-number-only duplicate test. The records relate to different suppliers and transactions. Invoice numbers must be checked together with Supplier_ID and other supporting fields before identifying duplicate payments. |

*Table 2: Investigation of a False-Positive Supplier Invoice Duplicate*

# Part C: Detailed Audit Risk Register

## Risk 1: Potential Duplicate Supplier Payment

| Required Element | Detailed Audit Assessment |
|---|---|
| Risk description and evidence | Power BI identified two payments of $32,000 against supplier S0012, invoice SUP5555: PAY01995 dated 2 May 2026 and PAY01996 dated 30 April 2026. Both were processed by U020 and approved by U025.<br>The $64,000 total may include an unauthorised duplicate payment of $32,000. |
| Relevant assertions and auditing standards | Occurrence and accuracy: Payments may have been recorded without a corresponding valid liability, potentially overstating expenses or understating cash.<br>ASA 315 requires identification and assessment of risks, ASA 330 governs audit responses, ASA 500 addresses sufficient appropriate audit evidence. |
| Internal control | • Preventive: The payment system should block repeated supplier-and-invoice combinations unless a documented exception receives independent approval.<br>• Detective: Accounts payable staff should review a weekly duplicate-payment exception report.<br>• |
| Test of control | • Population: Supplier payments during FY2026.<br>• Fields: Supplier_ID, Invoice_Number, Payment_ID, Payment_Amount, Processed_By and Approved_By.<br>• Procedure: Identify repeated supplier/invoice combinations and inspect the system's duplicate-payment controls and exception approvals.<br>• Expected: Duplicates are blocked or independently authorised.<br>• Exception: An additional payment without documented justification. |
| Substantive analytical procedure | Group supplier payments by Supplier_ID and Invoice_Number, comparing payment counts, amounts and dates. Identify invoices with multiple payments and investigate every instance where total payments exceed the authorised invoice amount. Flag identical payments made within seven days for detailed review. For supplier S0012, invoice SUP5555, two $32,000 payments were recorded two days apart. Compare these payments with the original supplier invoice, purchase order and payment records to determine whether the second payment was legitimate or |

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| | duplicated. |
| Substantive test of details | Select both SUP5555 payments. Inspect the supplier's original invoice, purchase order, approvals, bank statements and supplier statement.<br>Confirm whether two separate obligations existed and whether any excess payment was recovered.<br>Conclude whether the second $32,000 represents a valid payment, recoverable amount or misstatement requiring adjustment. |

*Table 3: Detailed Audit Risk Assessment — Potential Duplicate Supplier Payment*

## Risk 2: Post-Year-End Credit Notes and Revenue Cut-off

| Required Element | Detailed Audit Assessment |
|---|---|
| Risk description and evidence | Power BI identified $187.46K in credit notes issued during July 2026 relating to sales invoices dated between 1 July 2025 and 30 June 2026. These subsequent credits may indicate that revenue or receivables were overstated at year-end if the underlying conditions existed before 30 June 2026.<br>The credit notes are not automatically misstatements. |
| Relevant assertions and auditing standards | Cut-off, accuracy and valuation: Revenue must be recognised in the correct reporting period and receivables must be appropriately valued. ASA 315 addresses risk identification, ASA 330 requires appropriate audit responses, ASA 500 governs audit evidence and ASA 560 addresses subsequent events. |
| Internal control | • Preventive: Credit notes should require documented reasons and independent authorisation.<br>• Detective: Finance staff should review post-year-end credit notes against original invoice and dispatch dates before finalising the financial statements. |
| Test of control | • Population: Credit notes issued during July 2026.<br>• Fields: Credit_Note_ID, Original_Invoice_ID, Credit_Note_Date, Credit_Amount, Reason, Invoice_Date and Dispatch_Date.<br>• Procedure: Select credit notes relating to FY2026 sales and inspect approval evidence and the year-end review process.<br>• Expected: Each credit is authorised, supported and assessed for the correct accounting period.<br>• Exception: Missing approval, unsupported reason or |

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| | omitted cut-off assessment. |
| Substantive analytical procedure | Compare July 2026 credit notes totalling $187.46K with the average monthly credit notes for April–June 2026. Investigate any increase exceeding 25% of that average. Identify all full reversals after year-end and credit notes exceeding $10,000. Match these to original invoice dates, dispatch records and reasons for reversal to assess whether adjustments were required at 30 June 2026. For reused invoice IDs (INV14991–INV14995) verify the correct customer and transaction before drawing conclusions. |
| Substantive test of details | Select all individually material July credits and a sample of the remaining credits linked to FY2026 invoices. Inspect original invoices, dispatch records, return authorisations, customer correspondence and subsequent refunds. Determine when the underlying return or pricing issue arose.<br>Conclude whether a 30 June 2026 adjustment is required and quantify any identified misstatement. |

*Table 4: Detailed Audit Risk Assessment — Revenue Cut-off*

## Risk 3: Damaged and Obsolete Inventory Valuation

| Required Element | Detailed Audit Assessment |
|---|---|
| Risk description and evidence | Power BI identified inventory valued at $57.44M using quantity on hand multiplied by unit cost. Damaged inventory has an estimated recorded cost of $2.00M while inventory flagged as obsolete totals $227,460. These categories may overlap. If recoverable amounts are below recorded cost, inventory and profit may be overstated. |
| Relevant assertions and auditing standards | Valuation and allocation: Inventory should be recorded at the lower of cost and net realisable value under AASB 102. ASA 315 addresses risk assessment, ASA 330 audit responses, ASA 500 audit evidence and ASA 501 evidence concerning inventory existence and condition. |
| Internal control | • Preventive: Warehouse staff should identify and segregate damaged goods promptly.<br>• Detective: Management should review obsolete inventory, ageing reports and selling prices regularly.<br>• Corrective: Approved write-downs should be recorded when net realisable value falls below cost. |
| Test of control | • Population: Inventory items recorded for FY2026.<br>• Fields: Product_ID, Quantity_On_Hand, Damaged_Quantity, Obsolete_Flag, Unit_Cost, |

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| | Last_Count_Date and Review_Status.<br>• Procedure: Select damaged and obsolete items and inspect inspection records, management reviews and write-down approvals.<br>• Expected: Identified valuation issues are reviewed and appropriately adjusted.<br>• Exception: Damaged or obsolete items without documented assessment or approval. |
| Substantive analytical procedure | Compare each product's Unit_Cost with estimated net realisable value (NRV), using subsequent selling prices less estimated selling costs. Calculate potential write-downs for damaged and obsolete inventory, ensuring overlapping stock is not counted twice. Investigate all items where NRV is below unit cost, prioritising potential write-downs exceeding $10,000. Compare the results with management's existing inventory allowances and investigate material differences. |
| Substantive test of details | Select high-value damaged items, all individually material obsolete items and a sample of other inventory. Inspect physical condition reports, stock counts, subsequent sales invoices and estimated disposal costs.<br>Recalculate net realisable value and compare it with recorded cost.<br>Determine whether write-downs were required at 30 June 2026 and quantify any potential misstatement. |

*Table 5: Detailed Audit Risk Assessment — Damaged and Obsolete Inventory*

## Risk 4: Self-Approved Journal Entries

| Required Element | Detailed Audit Assessment |
|---|---|
| Risk description and evidence | Power BI identified 39 distinct journal entries where GLTable[User_ID] matches GLTable[Approver_ID]. These records indicate potential self-approval and a lack of independent journal review.<br>This increases the risk of unauthorised entries, management override and financial misstatement. |
| Relevant assertions and auditing standards | Occurrence, accuracy and classification: Unsupported journal entries could misstate account balances or expenses. ASA 240 addresses fraud and management override, ASA 315 covers risk assessment, ASA 330 requires appropriate audit responses and ASA 500 addresses audit evidence. |

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| Internal control | • Preventive: The accounting system should prevent users from approving their own journal entries.<br>• Detective: A finance manager independent of journal preparation should review exception reports and supporting documentation. |
| Test of control | • Population: All FY2026 general ledger journal entries.<br>• Fields: Journal_ID, User_ID, Approver_ID, Approval_Status, Posting_Date_Time and Amount.<br>• Procedure: Compare posting and approving user IDs and inspect approval logs for the 39 flagged journals.<br>• Expected: Independent approval is documented.<br>• Exception: Same-user approval without valid independent authorisation. |
| Substantive analytical procedure | Analyse all FY2026 journal entries by User_ID, Approver_ID, account, amount and posting date. Power BI identified 39 journals with matching posting and approving user IDs. Investigate all 39 entries using a zero-exception threshold for unexplained self-approval. Prioritise entries exceeding $50,000, round-dollar amounts and journals posted within 30 days of year-end. Compare monthly journal values by account with historical patterns to identify unusual fluctuations.<br><br>Examine supporting documents and approval records before concluding whether any entries represent financial misstatements or control deficiencies. |
| Substantive test of details | Examine all 39 flagged journals.<br>Inspect supporting invoices, management explanations, approval records and related account reconciliations. Trace material journal amounts to underlying evidence and assess whether they were recorded in the appropriate accounts and periods. Determine whether any entries require adjustment or indicate control deficiencies. |

*Table 6: Detailed Audit Risk Assessment — Self-Approved Journal Entries*

## Risk 5: Excessive User Access and Post-Termination Logins

| Required Element | Detailed Audit Assessment |
|---|---|
| Risk description and evidence | Power BI identified five users (U117, U051, U025, U018 and U013) who can both process and approve supplier payments.<br>Two users (U117 and U018) can also create suppliers.<br>Additionally, U117 and U118 have recorded login dates after |

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| | their employment termination dates. U117 holds supplier creation, payment processing, payment approval and journal posting permissions.<br>These findings indicate potential weaknesses in access management and segregation of duties. |
| Relevant assertions and auditing standards | Occurrence and accuracy: Inadequate access controls increase the risk of unauthorised transactions affecting financial statement balances. ASA 315 requires assessment of IT-related and internal control risks, ASA 240 addresses fraud risks, ASA 330 governs audit responses and ASA 500 addresses audit evidence. |
| Internal control | • Preventive: System access should follow role-based permissions with conflicting functions separated and terminated employees' access promptly disabled.<br>• Detective: Management should review user-access rights and login exception reports regularly. |
| Test of control | • Population: All user accounts and terminated employees during FY2026.<br>• Fields: User_ID, Role, Create_Supplier, Process_Payment, Approve_Payment, Last_Login and Termination_Date.<br>• Procedure: Compare user permissions against approved role profiles and inspect account deactivation records for terminated employees.<br>• Expected: Conflicting permissions are independently authorised, and terminated accounts are disabled promptly.<br>• Exception: Unauthorised conflicting permissions or a login after termination without a documented explanation. |
| Substantive analytical procedure | Analyse user permissions by role and compare them with actual payment and journal activity. Power BI identified five users with conflicting payment permissions, including two who can also create suppliers and two users with logins after termination. Calculate payment counts and values for these users and compare them with employees in similar roles.<br><br>Investigate every transaction processed and approved by the same user and every transaction occurring after an employee's termination date, using a zero-exception threshold. Review unusually high payment values and supplier activity linked to these accounts. Conflicting permissions indicate potential control weaknesses but do |

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| | not establish that unauthorised transactions occurred. |
| Substantive test of details | Investigate U117, U118 and the five users with conflicting payment permissions. Inspect HR termination records, access approvals, system login logs, supplier creation records, payment transactions and journal entries. Determine whether unauthorised activity occurred and trace any suspicious transactions to supporting documents. Conclude whether access restrictions failed and whether financial statement adjustments or additional audit procedures are necessary. |

*Table 7: Detailed Audit Risk Assessment — User Access and Post-Termination Logins*

# Part D: Interactive Power BI Report

**Page 1: Audit Overview**
The Audit Overview page presents NoLimit Retail Group Ltd's FY2026 operations, covering 400 customers, 250 suppliers and 12 branches. Key performance indicators include total net sales of $76.30 million, 4,995 unique sales invoices, $1.07 million in credit notes and $80.52 million in supplier payments. Interactive branch filtering, sales trends and credit note charts help reviewers compare performance and identify unusual transactions requiring further investigation.

**Page 2: Risks and Exceptions**
This page uses DAX measures to quantify potential audit risks. Potential Duplicate Supplier Invoices identifies repeated supplier invoice combinations while Payments to Unapproved Suppliers uses supplier approval records to flag 18 payments. Inventory measures use SUMX to calculate recorded inventory value and the cost of damaged or obsolete stock. Journal SoD Conflicts identifies 39 journals with matching posting and approval user IDs. Date-based filtering also identifies $187.46K in July 2026 credit notes relating to FY2026 sales.

**Page 3:  Audit Findings and Drill-through**
The final page summarises significant findings and provides transaction-level investigation. Reviewers can right-click an invoice visual and drill through to its credit note details including credit note ID, date, amount and reason. This supports tracing exceptions to individual records. Reused invoice IDs require additional verification before interpreting aggregated invoice values.

Together, the three pages support risk identification, investigation and audit planning.

# Part E: Memo to the Audit Partner

**To:** Audit Partner, BLG Partners
**From:** Senior Graduate Accountant
**Date:** 11 October 2026
**Subject:** NoLimit Retail Group Ltd — Preliminary Audit Risk Assessment for FY2026

Overall Risk Assessment
Our Power BI analysis highlighted several areas that need closer attention, particularly revenue cut-off, supplier payments, inventory valuation and journal approvals. We also found weaknesses in user-access permissions. These findings suggest that some controls may not be working as intended, so further testing is needed before we can determine whether the financial statements contain any misstatements.

Significant Audit Risks
Revenue cut-off (Risk 2): We identified $187,460 in credit notes issued in July 2026 that relate to FY2026 sales. This raises concerns about whether some revenue was recorded in the correct period. We need to check the original invoices, dispatch dates and reasons for the credit notes to determine whether adjustments were necessary at 30 June 2026. Five invoice IDs were also reused with different transaction details, so these records need further verification.

Supplier payments (Risk 1): We found two payments of $32,000 to supplier S0012 for the same invoice, SUP5555, made two days apart. This could be a duplicate payment so both transactions need to be checked against the original invoice, purchase order, approval records and bank statements. We also identified 18 payments totalling $800,791.56 to suppliers S0233 and S0241 which are currently marked as unapproved. Their approval status at the payment dates should be verified before concluding whether any controls were breached.

Inventory valuation (Risk 3): Our analysis shows total inventory of $57.44 million including $2 million in damaged stock and $227,460 in stock marked as obsolete. Some items may belong to both categories so we should not add these amounts together. We recommend checking stock condition reports, subsequent sales and estimated selling costs to determine whether any inventory needs to be written down at 30 June 2026.

Journal authorisation (Risk 4): We identified 39 journal entries where the same user appears to have posted and approved the transaction. This raises concerns about whether journal entries are being independently reviewed. We recommend checking the supporting documents and approval records for all 39 entries, particularly those with unusual amounts or posted near year-end. The related general ledger balances should also be checked against the trial balance.

Control Environment and Recommended Audit Response
Our review also found concerns with user-access controls (Risk 5). Five users can both process and approve supplier payments and two of them can also create suppliers. We also found that U117 and U118 had recorded login activity after their termination dates. We need to compare these findings with HR records, system logs and access approvals to check whether the permissions and activity were authorised.

Based on these findings, we recommend further testing of supplier payments, journal entries and other transactions linked to these users. Any significant control weaknesses should be discussed with management and where appropriate reported to those charged with governance. We should assess whether the findings affect the auditor's report only after obtaining sufficient evidence and considering their materiality.

**Conclusion:** The audit should continue with extra care and further checks in areas of concern before reaching a final conclusion.